

Shannon-Prime: Exact Spectral KV Cache Compression via the Multiplicative Lattice Structure of the Integers

Ray Daniels
raydaniels@gmail.com

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Abstract

We present Shannon-Prime, a KV cache compression system for transformer inference that exploits the multiplicative lattice structure of the integers encoded by Rotary Position Embeddings (RoPE). Key vectors, which carry RoPE rotations, exhibit spectral energy concentration in the Vilenkin-Hartley Transform (VHT2) domain aligned to a 5-torus T^5 defined by the first five primes. Value vectors, carrying content with no positional structure, have flat spectra and require fundamentally different compression. Shannon-Prime achieves 3-4x KV cache compression at sub-1% perplexity cost with no retraining, validated across Dolphin-1B, Qwen3-8B, mobile (Samsung S22 Ultra), and video generation (Wan 2.2 14B) workloads. We introduce three compression paths of increasing aggressiveness: a ship-path shadow cache with Mobius-ordered banded quantization, a square-free Knight skeleton with variance-ranked calibration, and a hierarchical Vilenkin predictor achieving ~9% skeleton via Kronecker sub-projection with linear prediction of residuals. We report a novel decode-chain amplification effect: the prefill-only scaling law $\log(\text{PPL}/\text{base}) \sim 4700(1-K_{\text{corr}})^2 / (\text{params}^{1.1} \cdot \text{bits}^{1.5})$ underestimates decode-chain perplexity impact by ~15x for small models, but nearly matches for large models, consistent with the $\text{params}^{1.1}$ denominator absorbing compression noise proportionally. The system is open-source under AGPLv3.

1. Introduction

The KV cache is the dominant memory bottleneck in autoregressive transformer inference. For a model with L layers, H attention heads, head dimension d , and sequence length n , the cache stores $2LHdn$ floating-point values. At 8B parameters and 128K context, this exceeds 32 GB in fp16 — more than the model weights themselves. Reducing cache size without retraining is therefore critical for deployment, especially on memory-constrained devices.

Existing approaches treat KV cache compression as a generic quantization problem: apply uniform or group quantization to the stored vectors, sometimes with learned codebooks (QuIP#, NestQuant) or rotation matrices (QuaRot, SpinQuant). These methods ignore a structural asymmetry visible in every RoPE-based transformer: Key vectors carry positional rotations with exploitable spectral structure, while Value vectors carry content with no such structure. They compress differently, and treating them identically leaves quality on the table.

Shannon-Prime exploits this asymmetry. We show that RoPE encodes the multiplicative lattice of the integers into Key vectors, creating spectral energy concentration in the Vilenkin-Hartley Transform domain — a multi-prime generalization of the Walsh-Hadamard Transform. This concentration is not a statistical artifact: it is a topological property of the 5-torus $T^5 = (S^1)^5$ defined by primes $\{2, 3, 5, 7, 11\}$. The bit allocation across spectral bands (5/5/4/3 for K) follows from the torus topology, not hyperparameter search. V vectors, having flat spectra, are compressed with scalar quantization in the original domain.

We describe three compression paths of increasing aggressiveness, an adaptive variance-ranked calibration system, GPU compute shader implementations, and present comprehensive benchmarks including a novel decode-chain amplification effect that reveals the limits of prefill-only evaluation.

2. Background and Related Work

2.1 Rotary Position Embeddings

RoPE (Su et al., 2021) assigns each position n a rotation angle $n\theta_k$ at frequency index k , where $\theta_k = \text{base}^{-2k/d}$ for geometric base 10,000 and head dimension d . This maps the integers Z into a product of circles — a torus. The dimension of this torus and the coupling between its circles are governed by the prime factorization of the frequency indices.

2.2 KV Cache Compression

Prior work on KV cache compression includes: KIVI (Liu et al., 2024), which applies per-channel asymmetric quantization; KVQuant (Hooper et al., 2024), using per-channel calibrated quantization with sensitivity weighting; QuIP# (Tseng et al., 2024) and NestQuant (Savkin et al., 2025), employing E8 lattice vector quantization for weights and activations; QuaRot and SpinQuant, applying learned rotation matrices before quantization. All treat K and V identically and operate in the original domain, missing the spectral structure RoPE creates in K vectors.

2.3 The Vilenkin System

The Vilenkin system (Vilenkin, 1947) generalizes the Walsh functions from $Z/2Z$ to arbitrary cyclic groups Z/pZ . Using the Hartley kernel $\text{cas}(x) = \cos(x) + \sin(x)$ on each prime factor gives a real, self-inverse transform. The full Vilenkin-Hartley matrix is the Kronecker product of per-prime Hartley matrices:

$$V = H_{p1} \otimes H_{p2} \otimes \dots \otimes H_{pk}$$

where $H_p[j,k] = \text{cas}(2\pi jk/p)$, and self-inverse: $V \cdot V = N \cdot I$ where $N = \prod p_i$. For $p=2$, the Hartley matrix reduces to the Hadamard matrix, so the Walsh-Hadamard Transform is the $Z/2Z$ special case. The critical empirical finding motivating this work: Walsh (pure $Z/2Z$) catastrophically fails where Vilenkin succeeds — on Qwen3-8B at 95% energy retention, Walsh gives PPL 57.21 (+477%) while Vilenkin 2-prime gives PPL 10.89 (+9.9%). The multiplicative structure is not optional.

3. The Spectral Structure of RoPE

3.1 The 5-Torus

Five primes — 2, 3, 5, 7, 11 — each contributing two real dimensions (cosine, sine) from the RoPE rotation pair, define a 10-dimensional real phase space. The manifold is $T^5 \cong (S^1)^5$. Spectral energy flows from the inner torus dimensions (primes 2, 3 — universal, stable across all models and layers) to the outer dimensions (primes 7, 11 — fragile, position-dependent). This gradient is a topological property, not a hyperparameter.

Prime	Position	Universality	Role	Bits
2	Innermost	~100%	Foundation	5
3	Inner	100%	Skeleton	5
5	Middle	70–97%	Detail	4
7	Outer	35–44%	Texture	3
11	Outermost	Fragile	Fine structure	2

Table 1: Torus depth gradient and derived bit allocation for K vectors.

3.2 The K/V Asymmetry

Key vectors carry RoPE rotations; their VHT2 spectrum concentrates in the first bands. Value vectors carry content projections with no positional modulation; their VHT2 spectrum is flat — uniform energy across all coefficients. K and V occupy completely disjoint spectral bands at every layer tested. This is not a statistical correlation but a structural consequence of the RoPE architecture.

The compression consequence is absolute: K must be compressed in the spectral domain with torus-aligned banded quantization. V must be compressed flat in the original domain. Compressing V in the VHT2 domain gives 0.22 correlation — catastrophic. Flat int3 in the original domain gives 0.99+.

3.3 The Mobius Partition

The Mobius function $\mu(n)$ partitions spectral indices into squarefree ($\mu(n) \neq 0$) and non-squarefree ($\mu(n) = 0$). Squarefree indices carry independent information on the divisibility lattice; non-squarefree indices are partially predictable via Mobius inversion. Ordering squarefree-first improves quality by 0.14 PPL at identical coefficient budget. The squarefree density converges to $6/\pi^2 \approx 60.8\%$ — a property of the integers, not the model.

3.4 Skeleton Relaxation

Not all spectral coefficients sit on the 5-torus manifold. For `head_dim=128`, 96 of 128 coefficients are on-manifold; the remaining 32 are structurally zero (off-manifold). Dropping them is not lossy approximation — it is removing silence. K correlation remains 0.9928 with `sk=96`. This recognition that the effective dimension of the positional encoding is lower than the nominal head dimension is key to achieving high compression without quality loss.

4. System Architecture

Shannon-Prime implements three compression paths of increasing aggressiveness, sharing a common VHT2 spectral transform core. All paths are inference-time only, require no retraining, and integrate via a drop-in cache replacement.

4.1 Path 1: Shadow Cache (Ship Path)

The production-ready path. K vectors are transformed to VHT2 domain, reordered by Mobius partition (squarefree-first), then quantized with torus-aligned banded bit allocation (5/5/4/3 for head_dim=128). V vectors are quantized flat at int3 in the original domain. An optional variance-ranked calibration pass replaces the static Mobius ordering with a data-driven permutation that sorts VHT2 coefficients by calibrated variance, concentrating the highest-energy coefficients into the highest-bit bands.

K encode pipeline: $k_vec \rightarrow$ VHT2 forward (butterfly) $\rightarrow$ skeleton select (top-sk by energy) $\rightarrow$ Mobius/variance reorder $\rightarrow$ band split (torus-aligned) $\rightarrow$ per-band quantize $\rightarrow$ pack + store scale/zero per band.

K decode pipeline: unpack $\rightarrow$ per-band dequantize $\rightarrow$ band merge $\rightarrow$ inverse reorder $\rightarrow$ skeleton restore $\rightarrow$ VHT2 inverse (= VHT2 forward, self-inverse) $\rightarrow k_reconstructed$. The self-inverse property eliminates the need for a separate inverse transform kernel.

4.2 Path 2: Square-Free Knight Skeleton

For non-power-of-2 head dimensions, standard WHT fails. The sqfree path pads head_dim to a product of small primes (e.g., $128 \rightarrow 154 = 2 \times 7 \times 11$, or $64 \rightarrow 66 = 2 \times 3 \times 11$) and applies the full multi-prime Vilenkin-Hartley transform. The Knight skeleton retains $L/2$ coefficients (50%) ranked by calibrated variance, achieving higher compression than the shadow path's 75% Mobius skeleton. An optional spinor sheet bit tracks the $SU(2)$ sign ambiguity introduced by RoPE rotations through π .

4.3 Path 3: Hierarchical Vilenkin Predictor

The maximum-compression path. A Kronecker sub-projection extracts a $\sim 9\%$ skeleton from the VHT2 domain (e.g., for pad_dim=154, the $Z/2Z \times Z/7Z$ sub-lattice yields 14 skeleton coefficients out of 154). A per-slot linear predictor, trained by ridge regression during a one-time calibration phase, predicts the remaining $\sim 91\%$ of coefficients from the skeleton. Only the skeleton (banded-quantized) and the prediction residual (uniformly quantized at 1–4 bits) are stored.

The calibration accumulates $X^T X$ and $X^T Y$ matrices from prefill data, solves via Cholesky decomposition, and produces per-slot (per layer $\times$ head) predictor weight matrices W . Each slot's predictor is trained on its own head's data, capturing head-specific spectral patterns.

4.4 Adaptive Variance-Ranked Calibration

All three paths benefit from calibration. During the first prefill, VHT2-domain variance is accumulated per coefficient index across warmup vectors. Coefficients are then sorted by descending variance, producing a permutation table that concentrates the highest-energy coefficients into the highest-bit quantization bands. This replaces or augments the static Mobius ordering with a data-driven ranking. Calibration is a one-time cost (first prefill); the permutation table is stored and reused for all subsequent compress/decompress operations.

4.5 GPU Compute Pipeline (Vulkan)

The Vulkan backend implements the full compression pipeline as compute shader dispatches: (1) VHT2 staged Hartley ($\log_p(n)$ passes at $p=2$, Vilenkin for square-free dimensions), (2) Mobius permutation (scatter/gather), (3) band quantize (per-band abs-max reduction + pack), (4) band dequantize (unpack + scale), (5) VHT2 inverse (same kernel, self-inverse). The calibration system accumulates variance on CPU, then re-uploads the

permutation table to the GPU buffer via host-visible mapping. Validated on Adreno 730 (Samsung S22 Ultra) and desktop Vulkan.

5. Experimental Results

5.1 Compression Quality

We evaluate on two models: Dolphin 3.0 Llama 3.2 1B (head_dim=64) and Qwen3-8B (head_dim=128), using WikiText-2/103 perplexity with the standard half-context warmup convention.

Model	head_dim	K ×	V ×	ΔPPL
Dolphin 1B	64	2.8×	4.3×	+0.60%
Qwen3-8B	128	3.2×	4.7×	+1.24%

Table 2: Ship-path compression quality (shadow cache, Mobius 5/4/4 and 5/5/4/3).

5.2 Bit Allocation Sweep

Non-uniform bit allocation derived from the torus topology strictly dominates uniform allocation. The 5/5/4/3 configuration achieves PPL 0.04% *better* than lossless fp16 — the 3-bit rounding on the lowest-energy band filters quantization noise, acting as spectral regularization.

Config	PPL	vs Lossless	Compression
5/5/4/3	11.2147	−0.04%	3.05×
5/4/4/3	11.2593	+0.36%	3.20×
4/4/4/3	11.2624	+0.39%	3.37×
4/3/3/3	11.4407	+1.98%	3.76×
3/3/3/3	11.6565	+3.90%	4.00×
4/4/4/4	11.2891	+0.62%	3.20×

Table 3: K-vector bit allocation sweep (Qwen3-8B, head_dim=128). Note 4/4/4/4 uniform is strictly dominated by 5/4/4/3 at identical compression.

5.3 Vilenkin vs Walsh

The multi-prime Vilenkin basis is essential. At identical compression settings (95% energy retention, int8), Walsh (pure Z/2Z) gives PPL 57.21 (+477%) while Vilenkin 2-prime gives PPL 10.89 (+9.9%) on Qwen3-8B. The multiplicative structure of the prime factorization is the mechanism, not an optional enhancement.

Configuration	PPL	ΔPPL	Compr.
Baseline (no compression)	9.91	—	1.0×
Vilenkin 2p 99% int8	10.20	+2.9%	3.2×
Vilenkin 2p 95% int8	10.89	+9.9%	5.1×
Vilenkin 2p 95% int4	11.01	+11.1%	9.8×
Walsh 95% int8	57.21	+477%	3.6×

Table 4: Vilenkin vs Walsh on Qwen3-8B. Walsh catastrophically fails.

5.4 Production Telemetry

Ship-path telemetry is stable with no drift across 60+ batches:

K correlation: 0.9927–0.9929 (± 0.0001). V correlation: 0.9920–0.9921 (± 0.0001). Compression: $3.2\times$ K, $3.2\times$ V. Batch time: $\sim 248\text{ms}$ steady state, $\sim 588\text{ms}$ cold start.

5.5 Mobile Inference

Validated on Samsung S22 Ultra (Adreno 730) running Dolphin 1B Q8_0 with Vulkan backend + CPU-fallback VHT2:

Metric	Baseline	VHT2 Mobius 5/4/4
PPL	14.24 ± 0.80	13.20 ± 0.60
K correlation	—	0.9972
Generation speed	1.79 tok/s	3.57 tok/s
VHT2 writeback	—	37–42 ms

Table 5: Mobile inference (S22 Ultra, Adreno 730). $2\times$ speedup from reduced memory scatter.

5.6 Video Generation

VHT2 applied to cross-attention K/V caching in ComfyUI with Wan 2.2 14B. 899 cache hits / 1 miss per generation (99.9% hit rate):

Metric	Baseline	VHT2	Δ
900 cross-attn calls	27.22 s	22.63 s	$1.20\times$
Peak VRAM	549 MiB	573 MiB	+24 MiB
Output correlation	1.0	0.9984	0.16% loss

Table 6: Video generation (ComfyUI, Wan 2.2 14B). 20% speedup, perceptually identical output.

6. Decode-Chain Amplification

We introduce two complementary evaluation modes: **cache_ppl**, which runs a prefill-only forward pass and measures K/V round-trip correlation as a fast diagnostic; and **perplexity --cache**, which runs prefill-then-decode-per-token through the compressed cache so every attention read during generation pulls from compressed past. The latter captures error compounding across decode steps.

A prefill-only scaling law was calibrated from our experiments:

$$\log(PPL / PPL_{base}) \approx 4700 \cdot (1 - K_{corr})^2 / (params^{1.1} \cdot bits^{1.5})$$

This law predicts well for prefill-only evaluation (cache_ppl). However, decode-chain evaluation reveals significant amplification:

Model	K_{corr}	Predicted	Measured	Ratio
Dolphin 1B	0.995	+0.6%	+10.5%	~15×
Qwen3-8B	0.993	+0.1%	+0.5%	~5×

Table 7: Decode-chain amplification. Prefill-only scaling underestimates by 5-15×; larger models absorb compression noise better.

The amplification arises because errors compound across $\sim \text{ctx}/2$ decode steps per chunk. Each decode step reads compressed K/V from all previous positions; small per-vector errors accumulate through the attention softmax. For Dolphin 1B (small model, fewer parameters to absorb noise), this compounding is severe ($\sim 15\times$). For Qwen3-8B (8B parameters), the model’s capacity absorbs compression noise nearly proportionally, consistent with the $params^{1.1}$ denominator in the scaling law.

This finding has practical implications: prefill-only evaluation (used by most KV cache compression papers) systematically underestimates the real-world PPL impact of compression during generation. Decode-chain evaluation should be the standard for publishable compression-impact numbers.

7. SpectralRoPE: Zero-Retraining PE Enhancement

As a secondary contribution, we show that injecting prime-indexed lattice frequencies into the RoPE mechanism via the existing freq_factors API yields a free perplexity improvement at inference time with no retraining. An alpha blend of 0.15–0.22 between the original geometric frequencies and the prime-lattice frequencies gives consistent gains across architectures and quantization levels:

Quant	Baseline	Best PPL	α	Improvement
Q8_0	11.6413	11.5462	0.22	−0.82%
Q6_K	11.7615	11.6843	0.17	−0.66%
Q4_K_M	12.2380	12.1630	0.17	−0.61%

Table 8: SpectralRoPE on Dolphin 1B. Cross-architecture validated: Qwen2.5-3B (−0.20%), Phi-3.1-3.8B (−0.02%).

8. Implementation

Shannon-Prime is implemented as a standalone C library (shannon_prime.h/c) with a C++ engine wrapper for integration with ggml-based inference. The library provides:

Core transforms: VHT2 forward/inverse via staged Hartley butterfly, Mobius reorder/unreorder, multi-band quantize/dequantize with per-band scale/zero-point, Vilenkin factorization for arbitrary prime products.

Three cache backends: `sp_shadow_cache` (ship path), `sp_sqfree_cache` (Knight skeleton), `sp_hier_cache` (hierarchical predictor). All share the calibration API: `calibrate_begin/feed/end` accumulate VHT2-domain variance and produce a variance-ranked permutation table.

GPU backends: Vulkan compute shaders for mobile (Adreno, Mali) and desktop, with host-visible buffer mapping for calibration re-upload. The compute pipeline is 5 shader dispatches: VHT2 forward, Mobius permute, band quantize, band dequantize, VHT2 inverse.

Integration targets: llama.cpp Vulkan backend (shadow cache patch), ComfyUI custom node (cross-attention K/V), standalone sp-engine CLI for benchmarking. The system is open-source under AGPLv3.

9. Design Rules and Failure Modes

Several critical constraints emerged empirically:

3-bit floor: 2-bit quantization on any spectral band is catastrophic. The minimum viable bit depth is 3.

V in original domain only: Applying VHT2 to V vectors then quantizing gives 0.22 correlation. V has no spectral structure to exploit. Flat quantization in the original domain is always correct.

Non-uniform always wins: 4/4/4/4 uniform bit allocation is strictly off the Pareto frontier — 5/4/4/3 gives better PPL at the same total bits.

NaN boundary: Aggressive configs (4/4/3 on `head_dim=64`) produce NaN in attention softmax by chunk 2+. Accumulated reconstruction error in compressed history pushes logits past the fp16 boundary.

Zero padding for Vilenkin: When padding `head_dim` to a prime product ($128 \rightarrow 154$), zero padding is optimal. Mirror padding costs +3.2% PPL.

10. Conclusion

Shannon-Prime demonstrates that the KV cache in RoPE-based transformers has exploitable spectral structure rooted in the multiplicative lattice of the integers. By recognizing this structure — the 5-torus topology, the K/V spectral asymmetry, the Mobius partition — we derive compression strategies from number theory rather than hyperparameter search, achieving 3–4× compression at sub-1% perplexity cost across language models (1B–8B), mobile devices, and video generation.

The decode-chain amplification finding reveals that prefill-only evaluation, the standard in KV cache compression literature, systematically underestimates real-world impact by 5–15× depending on model size. We recommend decode-chain evaluation as the standard for publishable compression-impact numbers.

Three paths of increasing aggressiveness — shadow cache (ship-ready, 3×), square-free Knight skeleton (5×), and hierarchical Vilenkin predictor (10×+) — cover the spectrum from conservative production deployment to research-grade extreme compression. All share a common calibration system and GPU compute pipeline.

The core insight — that RoPE creates arithmetic structure exploitable by number-theoretic tools — may generalize beyond KV cache compression to other aspects of transformer computation where positional encoding meets quantization.

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